

MA1001 — Calculus for Science Part 1

Winter 2023 Examination · Paper 1 · Full Marking Scheme

Module	MA1001 — Calculus for Science Part 1
Internal examiners	Dr. T. Carroll, Dr. S.J. Wills
External examiner	Prof. A. Belton
Head of School	Dr. K. Hayes
Institution	University College Cork
Session	Winter 2023, Paper 1
Duration	1.5 hours (+ 15 minutes reading time)
Structure	Answer all 6 questions . Marks for each part are shown on the paper: Q1 11, Q2 9, Q3 11, Q4 13, Q5 13, Q6 13 (70 in total).
Materials	Calculator allowed, but you must show where you used it. Formula sheet supplied with the paper (calculus, trigonometry, length/area/volume).

How to use this scheme. Every answer is worked out in small numbered steps, with a “Rules you need” box listing the rules used. Nothing is skipped, so you can see where every number comes from. Read the answer first, then follow the steps.

Addendum. The exam came with an addendum correcting Q1(iv) to “Sketch the graph of the function g...”. This scheme uses the corrected wording.

Calculators. Values that need a calculator are marked (*calculator*), because the paper says marks may be lost if you don't show where you used one.

Source basis. Every answer is built **only** from the MA1001 lecture notes and example/tutorial sheets: logarithms, graph sketching and transformations, derivative rules (product, chain, linear substitution), related rates, increasing/decreasing functions and the first derivative test, and limits at infinity. No textbook or outside material is used.

Mark splits. Part totals are the paper's own. The step-by-step split within each part is Study-Uni's suggested allocation.

Own-figure rule. Where a later step uses an earlier result, a correct method applied to a wrong earlier value still earns the method marks.

Study-Uni

Question 1

Piecewise functions, logarithms, graph transformations — 11 marks

Given: $f(x) = -2x$ if $x < -1$; $x^2 + 1$ if $-1 \leq x < 2$; $\ln x$ if $x \geq 2$. (Part (iv) uses the corrected wording from the exam addendum.)

1(i)

[2 marks]

What are the values $f(-1)$ and $f(2)$?

ANSWER

$$f(-1) = 2 \quad \text{and} \quad f(2) = \ln 2 \approx 0.693$$

WORKING — STEP BY STEP

1 How to read a piecewise function

f has three different rules. For any number x , check which condition x satisfies and use **only** that rule.

$$\text{Rule 1: } x < -1 \quad | \quad \text{Rule 2: } -1 \leq x < 2 \quad | \quad \text{Rule 3: } x \geq 2$$

2 Find $f(-1)$: which rule does $x = -1$ belong to?

Rule 1 needs $-1 < -1$: false. Rule 2 needs $-1 \leq -1$ **and** $-1 < 2$: both true. So use $x^2 + 1$:

$$f(-1) = (-1)^2 + 1 = 1 + 1 = 2$$

$((-1)^2 = (-1) \times (-1) = +1$: a minus times a minus is a plus.)

3 Find $f(2)$: which rule does $x = 2$ belong to?

Rule 2 needs $2 < 2$: false. Rule 3 needs $2 \geq 2$: true ($\geq$ allows equal). So use $\ln x$:

$$f(2) = \ln 2 \approx 0.693 \text{ (calculator)}$$

EXAM TECHNIQUE

The end points are the trap. $\leq$ and $\geq$ include the number itself; $<$ and $>$ don't. So -1 goes with $-1 \leq x < 2$, and 2 goes with $x \geq 2$.

Marks	Awarded for
1	$f(-1) = 2$ (Rule 2 used).
1	$f(2) = \ln 2 \approx 0.693$ (Rule 3 used).

1(ii)**[1 mark]**Is f increasing or decreasing on $(-\infty, -1)$?**ANSWER****Decreasing: on $(-\infty, -1)$, $f'(x) = -2$, which is negative.****RULES YOU NEED**

- f is decreasing where $f'(x) < 0$ and increasing where $f'(x) > 0$.

WORKING — STEP BY STEP**1 Which rule of f do we use here?**Every x in $(-\infty, -1)$ is less than -1 , so Rule 1 applies:

$$f(x) = -2x$$

2 Find the sign of the derivative f is **decreasing** where $f'(x) < 0$. The derivative of $-2x$ is just the number in front of x :

$$f'(x) = -2 < 0 \Rightarrow \text{decreasing}$$

Check with numbers: $f(-3) = -2 \times (-3) = 6$ and $f(-2) = -2 \times (-2) = 4$. As x goes up, f goes down.**Marks****Awarded for****1** $f'(x) = -2 < 0$, so f is decreasing.

1(iii)**[5 marks]**Sketch the graph $y = f(x)$.**ANSWER**

See the sketch: a straight line coming down to $(-1, 2)$, a U-shaped parabola with its lowest point at $(0, 1)$ rising to an open dot at $(2, 5)$, then a jump down to $(2, \ln 2)$ and a slowly rising $\ln x$ curve.

WORKING — STEP BY STEP**1 Left piece: $y = -2x$ for $x < -1$ (a straight line)**Table of values (just multiply by -2):

x	-3	-2	-1
$y = -2x$	6	4	2

It is a straight line with slope -2 , going **down** towards the point $(-1, 2)$.**2 Middle piece: $y = x^2 + 1$ for $-1 \leq x < 2$ (a parabola)**

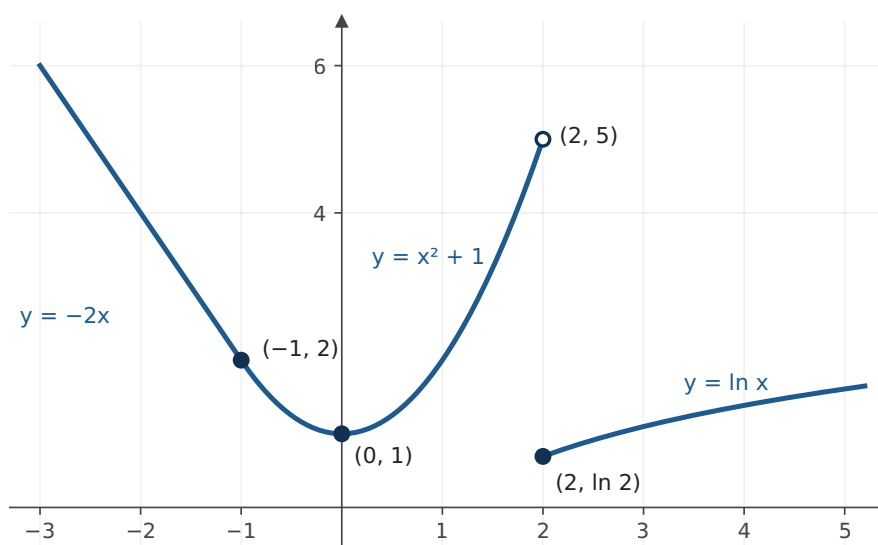
x	-1	0	1	1.5	2
$y = x^2 + 1$	2	1	2	3.25	5

For example $1.5^2 + 1 = 2.25 + 1 = 3.25$. It is the U-shaped graph of x^2 moved up by 1, with its lowest point at $(0, 1)$.**3 Dots at $x = -1$** The line reaches height 2 at $x = -1$ (open end), and Rule 2 gives $f(-1) = 2$ (filled). Both are at height 2, so the graph **joins up** here with no gap: one filled dot at $(-1, 2)$.**4 Dots at $x = 2$ (a jump)**Rule 2 has < 2 , so the parabola ends with an **open dot** at $(2, 5)$.Rule 3 has ≥ 2 , so the next piece starts with a **filled dot** at $(2, \ln 2) \approx (2, 0.69)$. The graph **jumps down** from 5 to about 0.69.

5 Right piece: $y = \ln x$ for $x \geq 2$

x	2	3	4	5
$y = \ln x$	0.69	1.10	1.39	1.61

(Calculator values.) $\ln x$ keeps increasing, but more and more slowly.



Marks	Awarded for
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- | | |
|---|---|
| 1 | Straight line $y = -2x$ for $x < -1$, coming down to $(-1, 2)$. |
| 2 | Parabola $x^2 + 1$ on $[-1, 2)$ with lowest point $(0, 1)$. |
| 1 | Open dot at $(2, 5)$ and filled dot at $(2, \ln 2)$: the jump at $x = 2$. |
| 1 | Increasing $\ln x$ curve for $x \geq 2$. |

1(iv)**[3 marks]**Sketch the graph of the function g , where $g(x) = f(-x) + 3$.**ANSWER**

$g(x) = 3 + \ln(-x)$ for $x \leq -2$; $x^2 + 4$ for $-2 < x \leq 1$; $2x + 3$ for $x > 1$ (see sketch).

WORKING — STEP BY STEP**1 What the two changes do to a graph**

$f(-x)$: put $-x$ in place of x . This flips the graph left-to-right (a mirror image in the y -axis).

$+ 3$: adds 3 to every y value, so the whole graph moves up by 3.

2 Work out $f(-x)$: replace x by $-x$ in each rule

Multiplying an inequality by -1 **flips its direction** (e.g. $-x < -1$ becomes $x > 1$).

$$\text{Rule 1: } -x < -1 \Leftrightarrow x > 1 \text{ gives } -2(-x) = 2x$$

$$\text{Rule 2: } -1 \leq -x < 2 \Leftrightarrow -2 < x \leq 1 \text{ gives } (-x)^2 + 1 = x^2 + 1$$

$$\text{Rule 3: } -x \geq 2 \Leftrightarrow x \leq -2 \text{ gives } \ln(-x)$$

3 Add 3 to each rule to get $g(x)$

$$g(x) = 3 + \ln(-x) \text{ if } x \leq -2$$

$$g(x) = x^2 + 4 \text{ if } -2 < x \leq 1$$

$$g(x) = 2x + 3 \text{ if } x > 1$$

4 Table of values

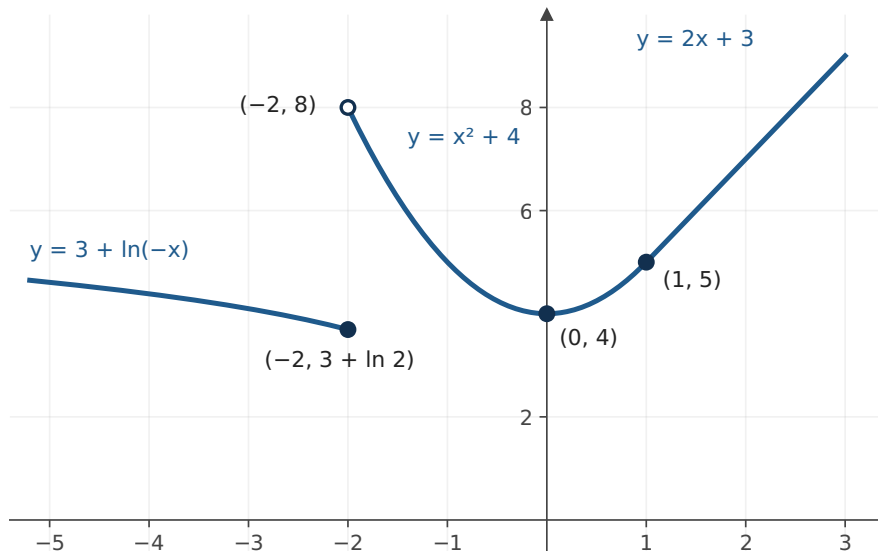
x	-4	-2	-1	0	1	2	3
$g(x)$	4.39	3.69	5	4	5	7	9

For example $g(-2) = 3 + \ln 2 \approx 3.69$, $g(0) = 0^2 + 4 = 4$ and $g(3) = 2(3) + 3 = 9$.

5 Dots at the break points

At $x = -2$: the first rule has $\leq$, so a **filled dot** at $(-2, 3 + \ln 2) \approx (-2, 3.69)$. The parabola has $<$, so it starts with an **open dot** at $(-2, (-2)^2 + 4) = (-2, 8)$. The graph **jumps up** here.

At $x = 1$: the parabola gives $1 + 4 = 5$ and the line gives $2 + 3 = 5$, so they **join up** at the filled dot $(1, 5)$.



EXAM TECHNIQUE

Do the mirror flip first, then move up by 3. The jump that was at $x = 2$ in part (iii) is now at $x = -2$.

Marks	Awarded for
1	$f(-x)$ worked out correctly (mirror image in the y-axis).
1	Shifted up by 3: $g(x) = 3 + \ln(-x)$, $x^2 + 4$, $2x + 3$ on the right intervals.
1	Sketch with the jump at $x = -2$ (filled $(-2, 3 + \ln 2)$, open $(-2, 8)$), lowest point $(0, 4)$, and the line from $(1, 5)$.

Question 1 Total: 11 Marks

Question 2

Logarithm rules, solving equations — 9 marks

2

[9 marks]

Solve the equation $\log_2(x - 5) = 4 - \log_2(x + 1)$.

ANSWER

$x = 7$ ($x = -3$ also comes out of the algebra but must be rejected)

RULES YOU NEED

- $\log A + \log B = \log(A \times B)$
- $\log_2 y = 4$ means $y = 2^4 = 16$
- You can only take the log of a positive number.

WORKING — STEP BY STEP

1 Find which x values are allowed

You can only take the log of a **positive** number:

$$x - 5 > 0 \Rightarrow x > 5 \quad \text{and} \quad x + 1 > 0 \Rightarrow x > -1$$

Both must be true, so:

$$x > 5$$

2 Get both logs on the same side

Add $\log_2(x + 1)$ to both sides:

$$\log_2(x - 5) + \log_2(x + 1) = 4$$

3 Combine the two logs into one

Use the rule: $\log A + \log B = \log(A \times B)$.

$$\log_2[(x - 5)(x + 1)] = 4$$

4 Remove the log

$\log_2(\text{something}) = 4$ means the something equals $2^4 = 2 \times 2 \times 2 \times 2 = 16$:

$$(x - 5)(x + 1) = 16$$

5 Multiply out and rearrange into a quadratic

$$(x - 5)(x + 1) = x^2 + x - 5x - 5 = x^2 - 4x - 5$$

Subtract 16 from both sides:

$$x^2 - 4x - 5 - 16 = 0 \Rightarrow x^2 - 4x - 21 = 0$$

6 Solve the quadratic

Find two numbers that **multiply to -21** and **add to -4**: these are -7 and $+3$ ($-7 \times 3 = -21$ and $-7 + 3 = -4$).

$$(x - 7)(x + 3) = 0 \Rightarrow x = 7 \text{ or } x = -3$$

7 Check both answers against Step 1

$x = -3$: $x - 5 = -8$, which is negative, so $\log_2(-8)$ doesn't exist. **Reject $x = -3$.**

$x = 7$: $x - 5 = 2$ and $x + 1 = 8$, both positive ✓. Check it in the original equation ($2 = 2^1$ and $8 = 2^3$):

$$\text{Left: } \log_2 2 = 1 \quad \text{Right: } 4 - \log_2 8 = 4 - 3 = 1 \quad \checkmark$$

So the only solution is **$x = 7$** .

EXAM TECHNIQUE

Always check every answer in the original equation. Combining the logs hides the rule that each log needs a positive input, so a fake answer can appear (here $x = -3$).

Marks	Awarded for
2	Conditions $x - 5 > 0$ and $x + 1 > 0$ stated (or both roots checked in the original equation).
2	Logs combined using $\log A + \log B = \log(AB)$ (or writing $4 = \log_2 16$).
2	Quadratic $x^2 - 4x - 21 = 0$.
1	Roots $x = 7$ and $x = -3$.
2	$x = -3$ rejected with a reason; final answer $x = 7$.

Question 2 Total: 9 Marks

Question 3

Product rule, chain rule, tangent lines — 11 marks

3

[11 marks]

Find the equation of the tangent line to the curve $y = x^2 \ln x - \sin(\pi x) + 2$ at the point on the graph where $x = 1$.

ANSWER

$$y = (1 + \pi)x + 1 - \pi \quad (\text{slope } 1 + \pi \approx 4.14)$$

RULES YOU NEED

- Tangent line: $y - b = m(x - a)$, where $m = dy/dx$ at the point (a, b) .
- Product rule: if $y = uv$ then $dy/dx = (du/dx)v + u(dv/dx)$.
- Derivatives: $\ln x \rightarrow 1/x$, and $\sin(ax) \rightarrow a \cos(ax)$. Also $\ln 1 = 0$, $\sin \pi = 0$, $\cos \pi = -1$.

WORKING — STEP BY STEP

1 What we need for a tangent line

A straight line needs **a point** (a, b) and **a slope** m . Then use:

$$y - b = m(x - a)$$

The slope of the tangent is the derivative dy/dx at that point.

2 Find the point (put $x = 1$ into the curve)

$\ln 1 = 0$ and, from the trig table, $\sin \pi = 0$:

$$y = 1^2 \times \ln 1 - \sin(\pi) + 2 = 1 \times 0 - 0 + 2 = 2 \Rightarrow \text{point } (1, 2)$$

3 Differentiate $x^2 \ln x$ using the product rule

It is two things multiplied. Name them and differentiate each:

$$u = x^2 \Rightarrow \frac{du}{dx} = 2x \quad v = \ln x \Rightarrow \frac{dv}{dx} = \frac{1}{x}$$

$$\frac{d}{dx}(x^2 \ln x) = \frac{du}{dx}v + u\frac{dv}{dx} = 2x \ln x + x^2 \cdot \frac{1}{x} = 2x \ln x + x$$

$(x^2 \div x = x.)$

4 Differentiate $\sin(\pi x)$ and the constant 2

$\sin(\pi x)$ is sin of $(a \text{ number} \times x)$. By the chain rule (linear substitution: the derivative of $f(ax + b)$ is $a \cdot f'(ax + b)$), with $a = \pi$:

$$\frac{d}{dx}(\sin(\pi x)) = \pi \cos(\pi x)$$

The derivative of a constant like 2 is 0.

5 Put the whole derivative together

Watch the minus sign in front of $\sin(\pi x)$:

$$\frac{dy}{dx} = 2x \ln x + x - \pi \cos(\pi x)$$

6 Find the slope at $x = 1$

$\ln 1 = 0$ and, from the trig table, $\cos \pi = -1$:

$$m = 2(1)(0) + 1 - \pi \times (-1) = 0 + 1 + \pi = 1 + \pi$$

7 Write down the equation of the line

Put $a = 1$, $b = 2$ and $m = 1 + \pi$ into $y - b = m(x - a)$:

$$y - 2 = (1 + \pi)(x - 1)$$

Multiply out and add 2 to both sides:

$$y = (1 + \pi)x - (1 + \pi) + 2 = (1 + \pi)x + 1 - \pi$$

In decimals: $y \approx 4.14x - 2.14$ (*calculator*).

EXAM TECHNIQUE

Use the trig table values $\sin \pi = 0$ and $\cos \pi = -1$ rather than the calculator. If you do use it, it must be in **radians**: in degree mode $\cos(3.14\dots)$ gives about 0.998, which is wrong.

Marks	Awarded for
2	Point (1, 2).
3	Product rule: derivative of $x^2 \ln x = 2x \ln x + x$.
2	Derivative of $\sin(\pi x) = \pi \cos(\pi x)$ (chain rule).
2	Slope $m = 1 + \pi$ at $x = 1$.
2	Tangent line $y - 2 = (1 + \pi)(x - 1)$, i.e. $y = (1 + \pi)x + 1 - \pi$.

Question 3 Total: 11 Marks

Question 4

Related rates of change — 13 marks

4

[13 marks]

A balloon is released at a point A that is 200 m away horizontally from an observer at point B. When the balloon is 150 m above the ground at point C it is rising with a speed of 3 m/s. At what rate is the angle of elevation θ changing for the observer? [Recall: $1 + \tan^2\theta = \sec^2\theta$.]

ANSWER

$d\theta/dt = 6/625 \approx 0.0096$ radians per second (the angle is increasing)

RULES YOU NEED

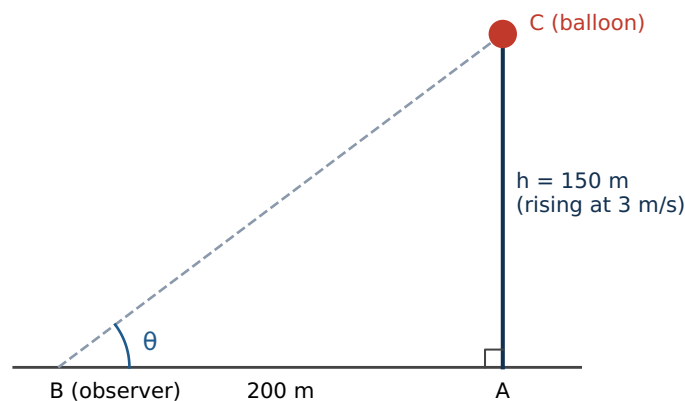
- SOHCAHTOA: $\tan \theta = \text{opposite} \div \text{adjacent}$.
- The derivative of $\tan \theta$ is $\sec^2\theta$; for a quantity that changes with time, $d/dt(\tan \theta) = \sec^2\theta \cdot d\theta/dt$.
- $1 + \tan^2\theta = \sec^2\theta$ (given in the question).

WORKING — STEP BY STEP

1 Draw the triangle and name the lengths

The balloon rises straight up from A, so B, A and C make a right-angled triangle with the right angle at A.

Let h = height of the balloon (the side AC). The side BA is always 200 m.



2 Write down what we know and what we want

$$\frac{dh}{dt} = 3 \text{ m/s} \quad \text{when} \quad h = 150 \text{ m}$$

We want $\frac{d\theta}{dt}$, how fast the angle is changing at that moment.

3 Link θ and h using SOHCAHTOA

From the observer at B: the **opposite** side is h and the **adjacent** side is 200. $\tan = \text{opposite} \div \text{adjacent}$:

$$\tan \theta = \frac{h}{200}$$

4 Differentiate both sides with respect to time t

The derivative of $\tan \theta$ is $\sec^2 \theta$, and θ changes with time, so the chain rule adds $d\theta/dt$ (like $d/dx(\sin y) = \cos y \cdot dy/dx$ in implicit differentiation):

$$\sec^2 \theta \cdot \frac{d\theta}{dt} = \frac{1}{200} \cdot \frac{dh}{dt}$$

5 Find $\sec^2 \theta$ at the moment $h = 150$

First $\tan \theta$, then use the hint $1 + \tan^2 \theta = \sec^2 \theta$:

$$\tan \theta = \frac{150}{200} = \frac{3}{4} \Rightarrow \sec^2 \theta = 1 + \left(\frac{3}{4}\right)^2 = 1 + \frac{9}{16} = \frac{25}{16}$$

6 Put in the numbers and solve

$$\frac{25}{16} \cdot \frac{d\theta}{dt} = \frac{1}{200} \times 3 = \frac{3}{200}$$

Divide both sides by $25/16$, which is the same as multiplying by $16/25$:

$$\frac{d\theta}{dt} = \frac{3}{200} \times \frac{16}{25} = \frac{48}{5000} = \frac{6}{625} \approx 0.0096 \text{ rad/s}$$

The answer is positive, so the angle of elevation is **increasing**, as you'd expect for a rising balloon.

EXAM TECHNIQUE

The rate of change of an angle is in **radians per second**. Don't swap h into the formula before differentiating: differentiate $\tan \theta = h/200$ first, then put $h = 150$ in.

Marks	Awarded for
3	$\tan \theta = h/200$ from the right-angled triangle.
4	Differentiating with respect to t : $\sec^2 \theta \cdot d\theta/dt = (1/200) dh/dt$.
3	$\tan \theta = 3/4$ and $\sec^2 \theta = 1 + 9/16 = 25/16$.
3	$d\theta/dt = 6/625 \approx 0.0096 \text{ rad/s}$.

Question 4 Total: 13 Marks

Question 5

Critical points, increasing and decreasing functions, curve sketching — 13 marks

Given: $f(x) = x^3 + 4x^2 - 3x + 4$

5 · Critical points

[6 marks]

Find all of the critical points for the function $f(x) = x^3 + 4x^2 - 3x + 4$.

ANSWER

Critical points at $x = -3$ and $x = 1/3$, i.e. the points $(-3, 22)$ and $(1/3, 94/27) \approx (0.33, 3.48)$

RULES YOU NEED

- Critical point: where $f'(x) = 0$.
- Power rule: the derivative of x^n is $n \cdot x^{n-1}$; a constant differentiates to 0.

WORKING — STEP BY STEP

1 What a critical point is

A critical point is where the slope is zero: $f'(x) = 0$. f is a polynomial, so f' exists everywhere.

2 Differentiate each term with the power rule

Bring the power down in front and take 1 off the power. The constant 4 differentiates to 0:

$$f'(x) = 3x^2 + 4(2x) - 3 + 0 = 3x^2 + 8x - 3$$

3 Solve $f'(x) = 0$ by factorising

$$3x^2 + 8x - 3 = 0$$

Multiply the first and last numbers: $3 \times (-3) = -9$. Find two numbers that **multiply to -9** and **add to 8**: 9 and -1. Split the middle term and group:

$$3x^2 + 9x - x - 3 = 3x(x + 3) - 1(x + 3) = (3x - 1)(x + 3) = 0$$

$$3x - 1 = 0 \Rightarrow x = 1/3 \quad x + 3 = 0 \Rightarrow x = -3$$

(The formula gives the same: $x = (-8 \pm \sqrt{(64 + 36)}) / 6 = (-8 \pm 10) / 6 = 1/3$ or -3 .)

4 Find the y-value at each critical point

Put each x back into the original $f(x)$:

$$f(-3) = (-3)^3 + 4(-3)^2 - 3(-3) + 4 = -27 + 36 + 9 + 4 = 22$$

$$f(1/3) = \frac{1}{27} + 4 \cdot \frac{1}{9} - 1 + 4 = \frac{1}{27} + \frac{12}{27} + \frac{81}{27} = \frac{94}{27} \approx 3.48$$

(Common denominator 27: $4/9 = 12/27$ and $-1 + 4 = 3 = 81/27$.)

Marks	Awarded for
2	$f'(x) = 3x^2 + 8x - 3$.
2	Factorised $(3x - 1)(x + 3) = 0$ (or formula), giving $x = -3$ and $x = 1/3$.
2	y-values $f(-3) = 22$ and $f(1/3) = 94/27$.

Study-Uni

5 · Increasing / decreasing**[4 marks]**

Find the intervals on which the function is increasing, and those on which it is decreasing.

ANSWER

**Increasing on $(-\infty, -3]$ and $[1/3, \infty)$; decreasing on $[-3, 1/3]$.
Local maximum at $x = -3$, local minimum at $x = 1/3$.**

WORKING — STEP BY STEP**1 The rule**

f is **increasing** where $f'(x) > 0$ and **decreasing** where $f'(x) < 0$.

2 Mark the split points and choose test points

The **split points** are the critical points $x = -3$ and $x = 1/3$. They cut the number line into three pieces. Pick one **test point** in each piece:

$$(-\infty, -3): \text{ test } x = -4 \quad (-3, 1/3): \text{ test } x = 0 \quad (1/3, \infty): \text{ test } x = 1$$

3 Work out $f' = 3x^2 + 8x - 3$ at each test point

$$x = -4: 3(16) + 8(-4) - 3 = 48 - 32 - 3 = 13 \rightarrow \text{positive}$$

$$x = 0: 0 + 0 - 3 = -3 \rightarrow \text{negative}$$

$$x = 1: 3 + 8 - 3 = 8 \rightarrow \text{positive}$$

4 Summary table

interval	$(-\infty, -3)$	$(-3, 1/3)$	$(1/3, \infty)$
sign of $f'(x)$	+	−	+
f is	increasing ↗	decreasing ↘	increasing ↗

5 Classify each critical point (first derivative test)

At $x = -3$ the graph goes up then down: a **hill**, so a local maximum, value 22.

At $x = 1/3$ it goes down then up: a **valley**, so a local minimum, value $94/27 \approx 3.48$.

Marks**Awarded for****3**Correct sign of f' on each of the three intervals.**1**

Intervals stated and critical points classified.

5 · Sketch**[3 marks]**Use all of this information to give a sketch of the graph $y = f(x)$.**ANSWER**

A cubic that rises to a hill at $(-3, 22)$, falls to a valley at $(1/3, 94/27)$, then rises for ever; it crosses the y-axis at $(0, 4)$ and the x-axis once, between -5 and -4 (see the sketch).

WORKING — STEP BY STEP**1 Plot the key points**

Hill (local max) at $(-3, 22)$. Valley (local min) at $(1/3, 94/27) \approx (0.33, 3.48)$.

Where it crosses the y-axis: put $x = 0$, $f(0) = 0 + 0 - 0 + 4 = 4$, so the point $(0, 4)$.

2 What happens at the far ends

For very large x , the x^3 term is much bigger than the others. So as $x \rightarrow \infty$, $f(x) \rightarrow \infty$ (up on the right), and as $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$ (down on the left).

3 Where does it cross the x-axis?

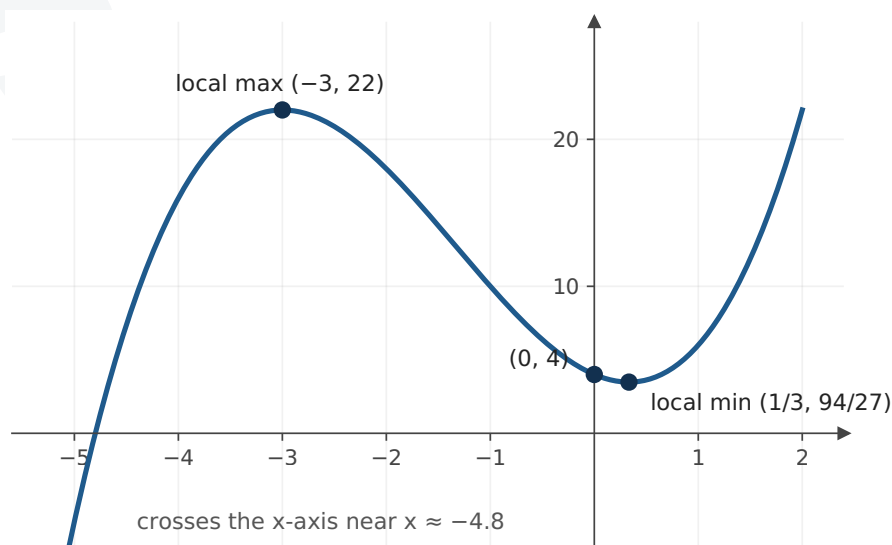
The valley at height $94/27$ is above the x-axis, so the graph only crosses the x-axis once, to the left of the hill. Try some values:

$$f(-5) = -125 + 100 + 15 + 4 = -6 \text{ (below)} \quad f(-4) = -64 + 64 + 12 + 4 = 16 \text{ (above)}$$

The sign changes, so it crosses somewhere between $x = -5$ and $x = -4$.

4 Join it all up

Come up from the bottom left, cross the x-axis between -5 and -4 , rise to the hill at $(-3, 22)$, fall through $(0, 4)$ to the valley at $(1/3, 94/27)$, then rise steeply.



Marks	Awarded for
1	End behaviour, y-intercept (0, 4), or the x-axis crossing used.
2	Sketch with maximum at $(-3, 22)$, minimum at $(1/3, 94/27)$, rising and falling in the right places.
Question 5 Total: 13 Marks	

Question 6

Differential equations, population model, limits at infinity — 13 marks

Given: $P(t) = \frac{3}{1 + ae^{-2t}} = 3(1 + ae^{-2t})^{-1}$.

6(i)

[6 marks]

Show that the function $P(t) = \frac{3}{1 + ae^{-2t}} = 3(1 + ae^{-2t})^{-1}$ satisfies the equation $\frac{dP}{dt} = 2P(1 - \frac{1}{3}P)$ for any choice of constant a .

ANSWER

Both sides work out to $\frac{6ae^{-2t}}{(1 + ae^{-2t})^2}$, so P satisfies the equation.

RULES YOU NEED

- Chain rule: $\frac{dP}{dt} = \frac{dP}{du} \times \frac{du}{dt}$.
- Power rule: the derivative of u^{-1} is $-u^{-2}$.
- The derivative of e^{kt} is $k e^{kt}$ (formula sheet).

WORKING — STEP BY STEP

1 The plan

Work out the **left side** dP/dt and the **right side** $2P(1 - P/3)$ separately, and show they are the same.

2 Left side: set up the chain rule

$P = 3(1 + ae^{-2t})^{-1}$ is a function inside a function. Let $u = 1 + ae^{-2t}$, so $P = 3u^{-1}$:

$$\frac{dP}{du} = 3 \times (-1)u^{-2} = -3u^{-2}$$

For u , the formula sheet says the derivative of e^{kt} is $k e^{kt}$. Here $k = -2$ (a is just a constant in front, and 1 differentiates to 0):

$$\frac{du}{dt} = a \times (-2)e^{-2t} = -2ae^{-2t}$$

3 Left side: multiply them together

Two minus signs make a plus:

$$\frac{dP}{dt} = \frac{dP}{du} \times \frac{du}{dt} = (-3u^{-2})(-2ae^{-2t}) = \frac{6ae^{-2t}}{(1 + ae^{-2t})^2}$$

4 Right side: work out $1 - P/3$

Dividing P by 3 cancels the 3 on top. Then write 1 as a fraction with the same bottom:

$$\frac{P}{3} = \frac{1}{1 + ae^{-2t}} \Rightarrow 1 - \frac{P}{3} = \frac{(1 + ae^{-2t}) - 1}{1 + ae^{-2t}} = \frac{ae^{-2t}}{1 + ae^{-2t}}$$

5 Right side: multiply by $2P$

$$2P(1 - \frac{P}{3}) = \frac{6}{1 + ae^{-2t}} \times \frac{ae^{-2t}}{1 + ae^{-2t}} = \frac{6ae^{-2t}}{(1 + ae^{-2t})^2}$$

6 Compare

Both sides are exactly $\frac{6ae^{-2t}}{(1 + ae^{-2t})^2}$, and we never needed to know what a is. So P satisfies the equation for **any** constant a .

Marks	Awarded for
3	$dP/dt = 6ae^{-2t}/(1 + ae^{-2t})^2$ using the chain rule.
2	Right side $2P(1 - P/3)$ simplified to the same expression.
1	Conclusion that the two sides are equal for any a .

6(ii)**[7 marks]**

The function P can be used to model population growth in ecology. If the initial population is 1, what is the population when $t = 0.5$? What is the eventual long-term trend for the population, i.e. what is $\lim_{t \rightarrow \infty} P(t)$?

ANSWER

$$a = 2; \quad P(0.5) = 3/(1 + 2e^{-1}) \approx 1.73; \quad \lim_{t \rightarrow \infty} P(t) = 3$$

RULES YOU NEED

- $e^0 = 1$.
- As $t \rightarrow \infty$, $e^{-2t} = 1/e^{2t} \rightarrow 0$.

WORKING — STEP BY STEP**1 Use the initial population to find a**

“Initial” means $t = 0$. Anything to the power 0 is 1, so $e^0 = 1$:

$$P(0) = \frac{3}{1 + a \cdot e^0} = \frac{3}{1 + a} = 1$$

Multiply both sides by $(1 + a)$, then subtract 1:

$$3 = 1 + a \Rightarrow a = 2$$

2 Write down P with $a = 2$

$$P(t) = \frac{3}{1 + 2e^{-2t}}$$

3 Find the population when $t = 0.5$

First the power: $-2 \times 0.5 = -1$. Then use the calculator for e^{-1} :

$$e^{-1} \approx 0.3679 \text{ (calculator)} \Rightarrow 1 + 2 \times 0.3679 = 1 + 0.7358 = 1.7358$$

$$P(0.5) = \frac{3}{1.7358} \approx 1.73 \text{ (calculator)}$$

4 Find the long-term trend ($t \rightarrow \infty$)

$e^{-2t} = 1/e^{2t}$. As t gets very large, e^{2t} becomes enormous, so 1 over it shrinks to 0:

$$e^{-2t} \rightarrow 0 \quad \text{as } t \rightarrow \infty$$

$$P(t) = \frac{3}{1 + 2e^{-2t}} \rightarrow \frac{3}{1 + 2 \times 0} = \frac{3}{1} = 3$$

5 What it means

The population starts at 1, grows (about 1.73 at $t = 0.5$), and levels off at **3** in the long run. It never goes above 3.

EXAM TECHNIQUE

The paper says to show where you used a calculator. Write “(calculator)” next to $e^{-1} \approx 0.3679$ and the final division.

Marks	Awarded for
3	$P(0) = 3/(1 + a) = 1$, so $a = 2$.
2	$P(0.5) = 3/(1 + 2e^{-1}) \approx 1.73$ (calculator use indicated).
2	$e^{-2t} \rightarrow 0$, so $\lim P(t) = 3$.

Question 6 Total: 13 Marks